

1.

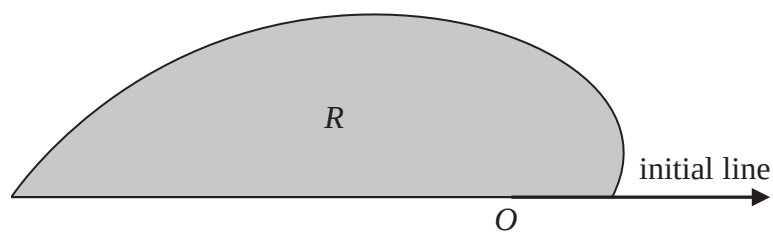
**Figure 1**

Figure 1 shows a sketch of the curve with polar equation

$$r = 2\sqrt{\sinh \theta + \cosh \theta} \quad 0 \leq \theta \leq \pi$$

The region R , shown shaded in Figure 1, is bounded by the initial line, the curve and the line with equation $\theta = \pi$

Use algebraic integration to determine the exact area of R giving your answer in the form $pe^q - r$ where p , q and r are real numbers to be found.

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA





Figure 1

Figure 1 shows the design for a table top in the shape of a rectangle $ABCD$. The length of the table, AB , is 1.2 m. The area inside the closed curve is made of glass and the surrounding area, shown shaded in Figure 1, is made of wood.

The perimeter of the glass is modelled by the curve with polar equation

$$r = 0.4 + a \cos 2\theta \quad 0 \leq \theta < 2\pi$$

where a is a constant.

- (a) Show that $a = 0.2$

(2)

Hence, given that $AD = 60$ cm,

- (b) find the area of the wooden part of the table top, giving your answer in m^2 to 3 significant figures.

(8)



3.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

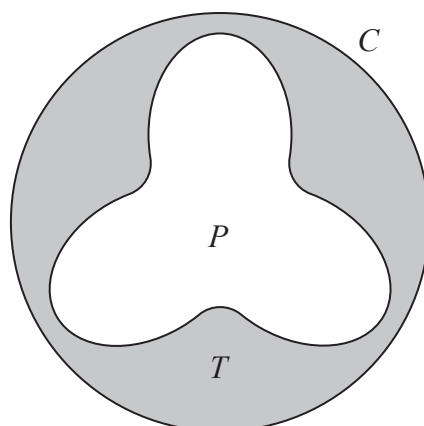


Figure 1

Figure 1 shows the design for a bathing pool.

The pool, P , shown unshaded in Figure 1, is surrounded by a tiled area, T , shown shaded in Figure 1.

The tiled area is bounded by the edge of the pool and by a circle, C , with radius 6 m.

The centre of the pool and the centre of the circle are the same point.

The edge of the pool is modelled by the curve with polar equation

$$r = 4 - a \sin 3\theta \quad 0 \leq \theta \leq 2\pi$$

where a is a positive constant.

Given that the shortest distance between the edge of the pool and the circle C is 0.5 m,

(a) determine the value of a .

(2)

(b) Hence, using algebraic integration, determine, according to the model, the exact area of T .

(6)



4. (a) Sketch the polar curve C , with equation

$$r = 3 + \sqrt{5} \cos \theta \quad 0 \leq \theta \leq 2\pi$$

On your sketch clearly label the pole, the initial line and the value of r at the point where the curve intersects the initial line.

(2)

The tangent to C at the point A , where $0 < \theta < \frac{\pi}{2}$, is parallel to the initial line.

- (b) Use calculus to show that at A

$$\cos \theta = \frac{1}{\sqrt{5}}$$

(4)

- (c) Hence determine the value of r at A .

(1)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



3.

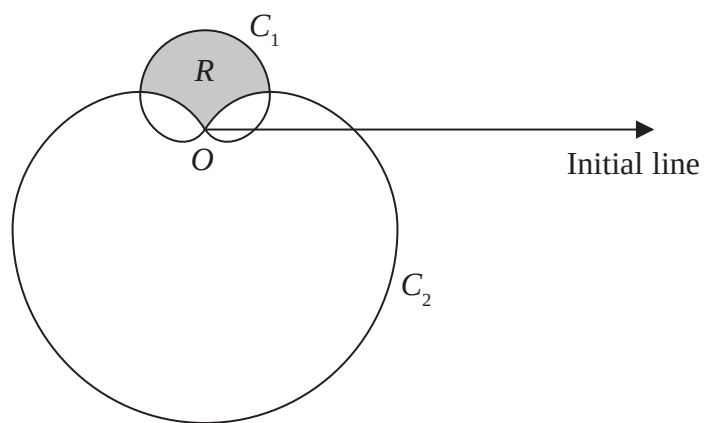
**Figure 1**

Figure 1 shows a sketch of two curves C_1 and C_2 with polar equations

$$C_1: r = (1 + \sin \theta) \quad 0 \leq \theta < 2\pi$$

$$C_2: r = 3(1 - \sin \theta) \quad 0 \leq \theta < 2\pi$$

The region R lies inside C_1 and outside C_2 and is shown shaded in Figure 1.

Show that the area of R is

$$p\sqrt{3} - q\pi$$

where p and q are integers to be determined.

(9)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA





$$r = 1 + \tan \theta \qquad 0 \leq \theta < \frac{\pi}{3}$$

(a) Use differentiation to prove that the polar coordinates of A are $\left(2, \frac{\pi}{4}\right)$

(b) Use calculus to show that the exact area of R is $\frac{1}{2}(1 - \ln 2)$ (6)